novamem

Tiered memory for AI agents · Apache-2.0 · self-hostable

azrtydxb.github.io/novamem

ONE MEMORY ACROSS EVERY AI AGENT YOU USE

The persistence layer your agents were already supposed to have.

LLMs forget every session. RAG silos every codebase. Each new agent re-discovers what the last one already learned. **novamem** is a single self-hostable service that gives Claude, ChatGPT, Cursor, Cline, Codex, and any other MCP- or HTTP-speaking agent a shared, durable memory — with hybrid retrieval, per-user isolation, and team-shareable sub-brains.

HOW IT WORKS — THREE RETRIEVAL SIGNALS, FUSED INTO ONE RANKED ANSWER

WARM — Postgres

Keyword-first hot path. Full-text search over every entry the user has ever written. Sub-millisecond. The default for "what did I just say."

COLD — Qdrant

Vector recall over older memory. Semantic similarity finds the entry whose words don't match but whose meaning does — the deep archive of long-decayed context.

GRAPH — FalkorDB

Adjacency from edges. Memories link to related memories. The graph surfaces context next-door — what was filed near this idea, on this project, by this collaborator.

WHAT YOU GET

- **Hybrid search by default.** One call returns warm + cold + graph signals fused with tunable per-signal weights.
- **Worthiness gate + dedup.** Trivial filler is rejected; near-duplicates merge — memory stays signal-dense.
- **Decay + dream cycle.** Stale entries fade; the daily compaction pass promotes graph edges between related memories.
- **Projects (sub-brains).** Per-user isolation by default, with shareable project scopes for team workflows.
- **Two transports, one engine.** JSON HTTP *and* MCP (Streamable HTTP + SSE) over the same store.
- **Built-in dashboard.** Sign-in, tokens, browse / today / graph / metrics — no third-party UI to wire up.

WHY YOU'D WANT IT

- **You switch agents constantly.** Claude, ChatGPT, Cursor, Cline, Codex — and each one starts the day with no idea what you did yesterday. novamem is the layer that remembers, so the agent doesn't have to.
- **Your team needs shared context.** A project sub-brain that on-call rotation, post-mortems, and design decisions all write into — every agent, every teammate, reads from the same place.
- **Long-running investigations.** A research project that spans months and many conversations needs a memory that doesn't get truncated when the context window fills up.
- **You don't want SaaS.** Self-hostable, Apache-2.0, your data stays on your infra. Same code from a laptop Compose stack to a multi-tenant Kubernetes deploy.

THREE DEPLOYS — SAME PRODUCT, DIFFERENT SCALE

LAPTOP SOLO

Docker Compose, single user, no auth. Stand it up with one command and connect every host on your machine via the `novamem-init` CLI.

HOMELAB TEAM

Self-hosted Kubernetes / k3s. Multi-user with Better Auth, per-user bearer tokens, project sharing, and the admin dashboard for tenant + token CRUD.

MULTI-TENANT ORG

Same code, multi-tenant mode. Per-tenant isolation, audit log, Prometheus metrics, JWT/JWKS for service-to-service. Run a memory backbone for a whole company.

STACK

Fastify · TypeScript · Postgres · Qdrant · FalkorDB

TRANSPORTS

HTTP/JSON · MCP `Streamable` · MCP `SSE`

AUTH MODES

`bearer` · `user` (Better Auth)

Give your agents a memory.

Self-hostable in minutes. Apache-2.0. No SaaS dependency. No vendor lock-in.

docs azrtydxb.github.io/novamem
code github.com/azrtydxb/novamem
npm @azrtydxb/novamem-init
img ghcr.io/azrtydxb/novamem